

# TalentPulse

AI-Powered Hiring Intelligence for GTM & Recruiting Teams

**16,611**

Live LinkedIn Jobs

**15**

Bright Data Snapshots

**9**

AI Role Categories

Bright Data × lablab.ai Hackathon 2026 | Solo Builder: Shriyutha Udayashankar | Track 1: GTM Intelligence | Track 2: Finance & Market Intelligence

**BRIGHT DATA**

# The Problem

## GTM teams are flying blind on talent market shifts.

Manual job board searches — hours of copy-pasting

LinkedIn Talent Insights costs \$10,000+/year for monthly reports

Lightcast / Burning Glass = \$50,000+/year for stale data

Missing real-time signals — by the time you know, it's too late

No way to know NVIDIA added 129 roles this week without expensive tooling

# The Solution: TalentPulse

## Bright Data Scraping

15 snapshots · 16,611 live LinkedIn jobs · USA + India

## Claude Agentic Loop

4 autonomous tool calls · market intelligence brief · delta alerts

## ML Salary Predictor

Random Forest  $R^2=0.5139$  · 63 skills · TF-IDF features

## GTM Scoring

6,716 companies ranked by buying intent · NVIDIA #1 (107)

## Real-Time Delta Detection

NVIDIA +129 roles this week · Cymertek 63→0 (freeze signal)

## Statistical Proof

T-tests + ANOVA · PyTorch = +\$25K premium ( $p<0.001$ )

# Key Findings

**\$200K**

SF Median Salary

AI Engineer

**26.7%**

LLMs/GenAI Demand

Up from ~0% in 24m

**+\$25K**

PyTorch Premium

$p < 0.001$  \*\*\*

**107.1**

NVIDIA GTM Score

#1 Buying Intent

NVIDIA #1 GTM Target — score 107.1 · 163 open AI roles · GPU infrastructure buildout

Capital One #2 — score 105.1 · 187 roles · financial services competing with Big Tech

Stakeholder management +5.7% · market shifting from BUILD → DEPLOY & GOVERN

NVIDIA added 129 net new roles in ONE WEEK — detected automatically by TalentPulse

Cymertek: 63 → 0 roles — possible hiring freeze or acquisition signal

# Bright Data Integration

BRIGHT DATA

## USA Snapshots (8)

|                |            |
|----------------|------------|
| Data Scientist | 4,467 jobs |
| ML Engineer    | 6,423 jobs |
| AI Engineer    | 1,982 jobs |
| LLM Engineer   | 1,157 jobs |
| NLP Engineer   | 958 jobs   |
| Data Engineer  | 5,937 jobs |
| Data Analyst   | 334 jobs   |
| Biz Analyst    | 334 jobs   |

## India Snapshots (7)

|                |          |
|----------------|----------|
| Data Scientist | 144 jobs |
| ML Engineer    | 64 jobs  |
| AI Engineer    | 41 jobs  |
| Data Engineer  | 91 jobs  |
| NLP Engineer   | 27 jobs  |
| LLM Engineer   | 25 jobs  |
| Data Analyst   | 153 jobs |

Total: 16,611 unique jobs | 15 snapshots | 9 role categories | 2 countries

# Claude Sonnet 4.6 — Agentic Loop

1

**analyze\_city(San Francisco)**

→ \$200K median salary · 2,376 jobs

2

**analyze\_skill(IIm\_genai)**

→ +\$25K premium · 26.7% demand

3

**get\_top\_gtm\_targets(top\_n=5)**

→ NVIDIA #1 (107) · Capital One #2 (105)

4

**find\_hot\_opportunities(score=80)**

→ 2,429 hot jobs · low competition detected

*Claude autonomously decides what to analyze — no hardcoded logic. This is genuine agentic behavior.*

# Tech Stack

## Data

Bright Data LinkedIn Scraper API · 15 snapshots · USA + India

## AI

Claude Sonnet 4.6 · Agentic loop · 4 autonomous tool calls

## ML

Random Forest · Gradient Boosting · TF-IDF ·  $R^2=0.5139$

## Stats

T-tests · ANOVA ( $F=75.07$ ) · Pearson correlation ·  $p<0.001$

## Analysis

Python · Pandas · NumPy · Scikit-learn · SciPy

## Dashboard

Streamlit · Plotly · Matplotlib · Seaborn

# Live App & Demo

## Live App URL

[talentpulse-nw2wgszdfsrasc5vre28dr.streamlit.app](https://talentpulse-nw2wgszdfsrasc5vre28dr.streamlit.app)

## GitHub Repository

[github.com/shriyutha/TalentPulse](https://github.com/shriyutha/TalentPulse) — includes all 15 Bright Data snapshot IDs + full analysis notebook

## 6 Dashboard Tabs:

### Skills

Top 20 skills ranked by demand

### Companies

GTM scores for 6,716 companies

### Opportunities

Hot jobs by competition score

### Salary

Distribution + seniority breakdown

### AI Brief

Claude market intelligence report

### Delta Alerts

Real-time market change signals